

Aims 1 and 2: Findings Brief

Measuring the multilinguality of LLMs: ten findings

(methods, statistics, and the error ledger are in the accompanying technical report)

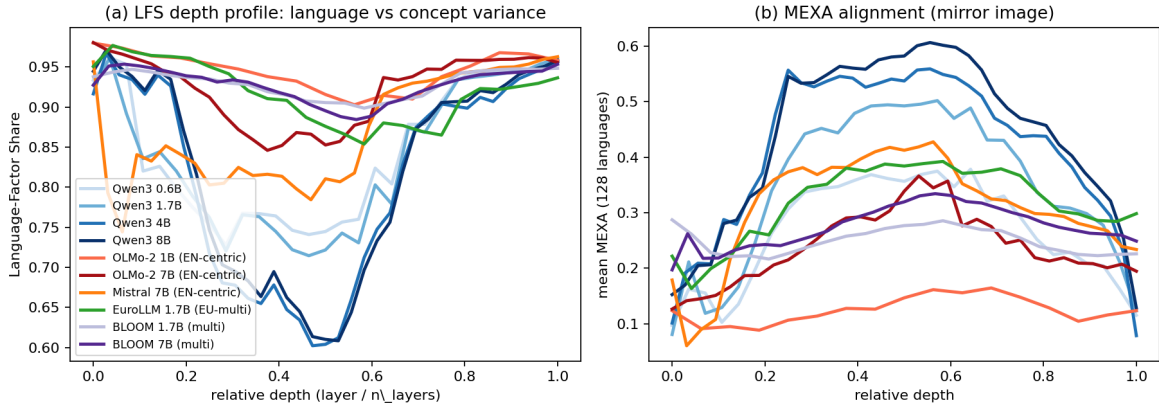
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The assignment (proposal §3.1.4): build an intrinsic metric of LLM multilinguality. **The contribution:** one primary metric, the **Language-Factor Share** (LFS); the share of systematic representation variance attributable to language rather than shared concept structure, an exact two-way variance decomposition on a balanced translation grid; with a validation methodology (“confounder-adjusted validity”) that removes observable confounds before any claim is made. Retrieval alignment, tail analysis (Alignment-at-Risk), downstream performance, and generation-time context use serve as *validation instruments*, not additional definitions of multilinguality. **Scale:** 17 open models (0.6B–9B), 122–128 languages, all claims either pre-registered or confound-controlled.

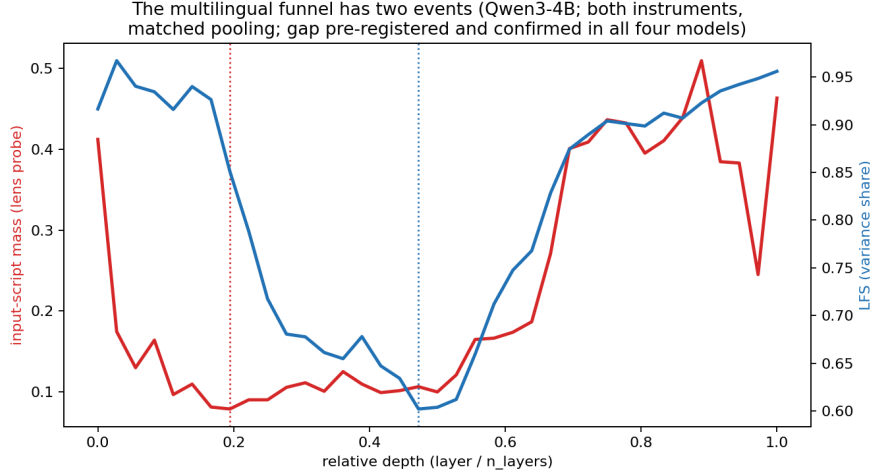
Finding 1: Your §3.1.1 hypothesis, measured

The proposal asks “how strong the language component in the representation is.” The **Language-Factor Share** (LFS) is a direct operational measure of it, per layer: every model shows language-specific → shared → language-specific processing, and the depth of the shared dip varies almost $10\times$ across models (0.04–0.34). By construction, LFS measures the *additive, systematic* language share of representation variance; language-specific rotations or concept-specific distortions fall outside it. A pre-registered falsification test suite has since *measured* that boundary: rotation is 0.06–0.2% of real cross-language misalignment (offset+scale: 63–97%), it does not predict performance beyond observables, and the one gaming attack that is missed by every alignment metric tested (global collapse; MEXA included) is caught exactly by a validated variance-preservation check.



Finding 2: The depth profile has two events, not one

Pre-registered and confirmed twice: models abandon the input language’s *script* early ($\sim 5\text{--}20\%$ depth) but *meanings* converge only mid-network ($\sim 30\text{--}50\%$). Surface form and semantics run on different schedules.



Finding 3: Family separates more than scale (in this grid)

Scale deepens the shared space within a family (Qwen3: dip 0.21 $\rightarrow$ 0.34; low-resource alignment $\times 6$), but between-family differences dwarf the within-family trend (“family” bundles data mixture, token budget, tokenizer, and curriculum; an observational contrast, not a causal claim): Qwen3-1.7B out-converges every other lab’s 7B. **BLOOM (deliberately 46-language) has the shallowest concept space of all, unchanged from 1.7B to 7B:** its alignment concentrates at the embedding layer (surface form, not meaning). *Coverage is a data property; convergence is a training property.* For Aim 2 this is the license: the quantity is set by training choices, not an automatic gift of scale.

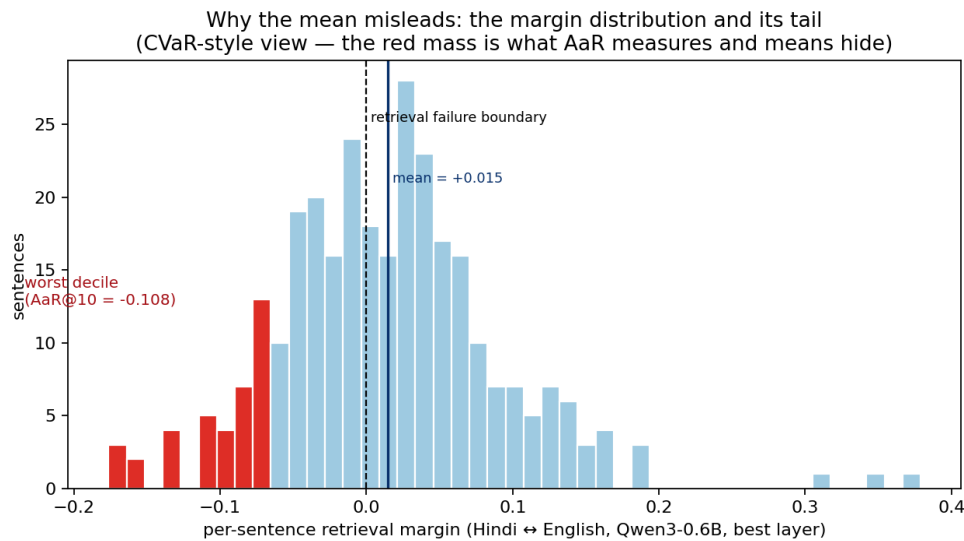
Finding 4: “Thinks in English” is resource-graded and fading

A multilingual latent factor predicts each language’s representations better than English does (113 to 124 of 127 languages depending on the model, 124 for most; confound-controlled; a modest $+0.08 R^2$ after the smoothness correction). English is the best single bridge for high-resource targets (9/10) but almost never for low-resource ones (1/8, where Swahili, Hindi, and Indonesian mediate regionally), and its hub share *falls* with scale (45 $\rightarrow$ 25 of 127 across Qwen3 sizes). Bonus for §3.1.2: mid-layer states reach max cosine ~ 0.10 to *any* output embedding; the Logit Lens is weakest exactly where the literature applies it.

Finding 5: The mean hides the tail, and the tail is semantic

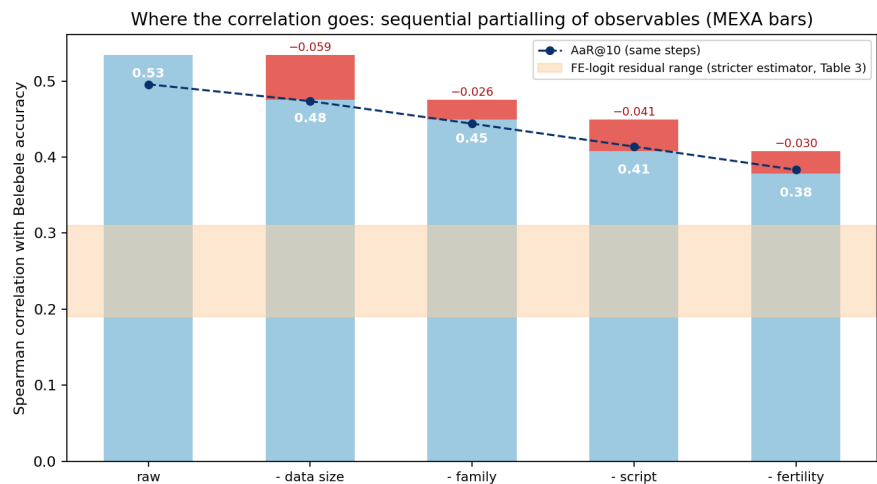
Mid-resource languages pass average-based checks while their worst-decile sentences sit at retrieval failure. Measure those sentences: named entities and numerals are alignment *anchors*; what fails is discourse-dependent meaning (“Interest remained high after the hearing”), culturally-bound vocabulary, and crowded boilerplate. **Alignment-at-Risk** (AaR) reports this tail; means cannot. At the 1,500-sentence scale ($n=1500/\text{language}$) this is the *most common* condition, not an anecdote.

dote: 40–50% of languages in the strongest models are mean-positive but tail-negative (Qwen3-8B: Turkish and Greek carry means of +0.13 over worst-percentile margins of −0.26 and −0.24).



Finding 6: Confounder-adjusted validity: on our grid, the standard validation number is ~40–60% confound

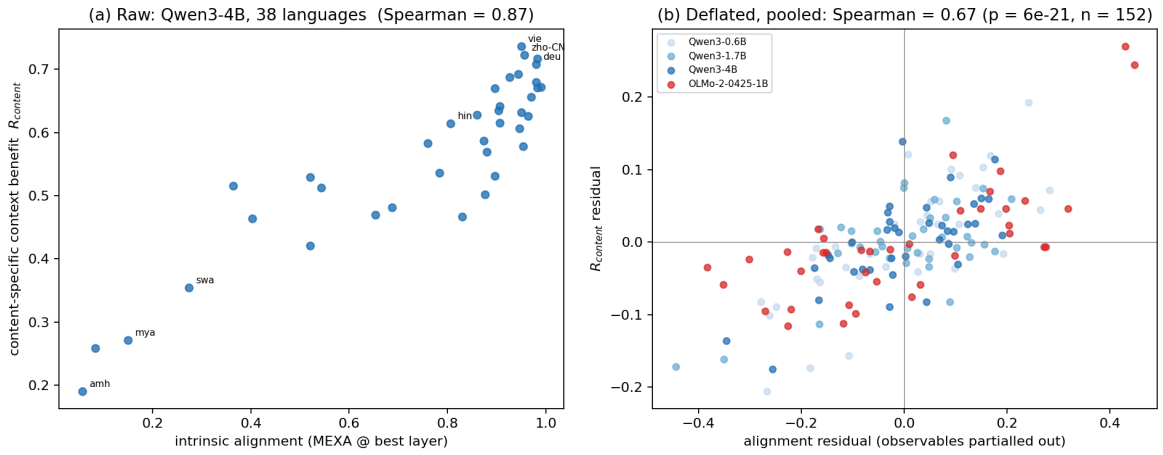
Intrinsic metrics are validated by raw correlation with per-language performance; but data-rich languages score high on both. Partialling out data size, family, script, and tokenizer cost: the confounder-adjusted residual range is $\rho \approx 0.19\text{--}0.31$ (raw: 0.50–0.53). Real signal survives (the metrics are not merely token-counting), but on this model–language grid and benchmark, the raw numbers are roughly half confound. (Applied to our own best number too; see Finding 7.) The July figure of about 0.59 on a benchmark-capable subset is withdrawn: it is not reproducible from any stored artifact, and a re-derivation gives 0.487 [0.365, 0.554].



Finding 7: The shared-space measurements predict behavior beyond observables

Pre-registered: per-language alignment, measured at the layer the variance decomposition identifies, predicts how much a model actually *uses* a foreign-language context sentence during generation (mismatched-context controlled, 40 languages). The chain: LFS locates the shared region; alignment in that region predicts behavior; the structure LFS finds is behaviorally meaningful. Confounder-adjusted by the same machinery as Finding 6: pooled residual $\rho = 0.67$ (language-clustered CI [0.55, 0.77]; strictest per-language treatment $\rho = 0.46$, $p = 0.004$).

This chain is observational, and it has now been tested under intervention. A six-condition training study at three seeds directly optimized LFS downward by five different objectives. In every condition, at every seed, tail alignment moved down with it (Pearson +0.924 across eighteen condition-seed points), the opposite of the association seen across pretrained models. The association therefore does not transfer to these interventions. That is a limit on *use*, not a refutation of the measurement: LFS is a measurement and is not a training target.



Finding 8: Risk-adjusted model comparison

Mean accuracy and worst-decile accuracy give different answers; the mean-versus-tail view makes dominance visible immediately, and exposes a hard floor: ~ 30 languages have no usable alignment in *any* open model under 8B; and 28 of 122 Belebele languages are absent from both major public web-corpus censuses.

Aim 1 built the instrument; Aim 2 established what it must not be used for, and what its summary statistic must not hide.

How these claims were validated: every main claim was either pre-registered before running or adjusted for observables. The pre-registration ledger totals **26 predictions, 18 passed:** 13 experimental predictions across two rounds (9 passed; 4 failed, all productive), 4/4 blind prospective predictions, and a 28-condition Aim-2 study scored against pre-committed criteria (5 word-level predictions, 4 passed). **Ten** self-caught errors are documented with mechanisms in the full report’s ledger. Code, data pipeline, and protocols are release-ready.